

Cheat Sheet: Related Rates vs. Optimization

Why these two get confused

Both topics show up as word problems with geometry (rectangles, cones, ladders, boxes), both use derivatives, and both can sound similar on a quick read. But they're asking fundamentally different questions, and mixing up the approach wastes a lot of time. This sheet is a fast way to sort a problem into the right bucket before you start solving.

The one-question test

Ask: is something changing over time, with a rate already given to you?

- Yes, and I need to find a different rate at a specific instant → Related Rates
 - No — I just need to find the single best (biggest/smallest/cheapest/optimal) value, no rates involved → Optimization
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Related Rates — the signals

- The problem gives you at least one rate already (a $\frac{dx}{dt}$ or $\frac{dr}{dt}$, phrased as “increasing at,” “decreasing at,” “moving at”).
- It asks “how fast is [something] changing” or “at what rate.”
- Multiple quantities are all changing at the same time, tied together by a geometric or physical equation.
- The final answer is a rate — it has “per unit time” in its units (ft/s, gal/min, rad/s, etc.).

Example phrasing: “A ladder is sliding down a wall... the bottom is moving away at 2 ft/s... how fast is the top sliding down when...”

The approach: find an equation connecting the variables, differentiate both sides with respect to t (implicit differentiation in disguise), then plug in the specific instant's values — never plug in numbers before differentiating.

Optimization — the signals

- The problem asks you to maximize or minimize something — biggest area, smallest cost, least material, greatest volume, highest profit.
- There's usually a constraint (a fixed amount of fencing, a fixed budget, a fixed volume) tying two variables together.
- Nothing is described as changing over time — you're looking for one best, unchanging value.
- The final answer is a value — a dimension, a price, an amount — not a rate.

Example phrasing: “A farmer has 400 m of fencing... find the dimensions that give the largest possible area.”

The approach: write the objective as a function of one variable (using the constraint to eliminate the other variable), differentiate, set the derivative to 0, and confirm it's actually a max or min.

The trap: when “time” language shows up in an optimization problem

Some optimization problems still mention t (time) as the single variable — for example, “find the time t that minimizes cost.” This is still optimization, not related rates, because there’s only one quantity varying (cost, as a function of t), not multiple linked rates. The test still applies: are you given a rate and asked for another rate (related rates), or are you finding one best value (optimization)? If it’s the second, treat t just like any other variable and optimize normally.

Side-by-side comparison

	Related Rates	Optimization
What’s given	A known rate at some instant	A fixed constraint (budget, material, etc.)
What’s asked	A different rate, at that same instant	The single best value of some quantity
Key tool	Implicit differentiation w.r.t. t	Setting $f'(x) = 0$, then confirming max/min
When to plug in numbers	After differentiating, never before	After finding the critical point
Answer’s units	Rate (per unit time)	A plain value (length, dollars, etc.)